Compatible Jacobi Endpoints and Quantitative Pole Control for Global Support under Weak MTW

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Abstract

The assigned manuscript asserts global supporting mountains and convex injectivity domains under weak Ma–Trudinger–Wang curvature, allowing conjugate cut points. We isolate its actual Jacobi endpoint problem and retain the Riemannian metric, time normalization and minimizing hypothesis. At a conjugate endpoint, solvability requires orthogonality to the initial Jacobi kernel; otherwise the affine index problem is unbounded below. For proper radial contractions we prove a rank-resolved pole formula with an explicit bounded remainder, using a nonsingular rescaled endpoint matrix rather than an inverse at conjugacy. The same matrix gives a conditional error enclosure, and the pole formula identifies the precise square-root scale of admissible kernel contamination. Compatible terminal data admit an energy enclosure only with a verified positive complement gap. These are local consequences of classical Jacobi, Fredholm and perturbation arguments; the source's MTW continuation, arbitrary-potential collision argument and globalization remain separate inputs. Contribution assessment: C3/5, a quantitative rank-resolved endpoint and contamination certificate strengthens the source's local singular-Hessian analysis without certifying or extending its global theorem.

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1 The source problem and its logical boundaries

Let (M, g) be a smooth connected compact Riemannian manifold without boundary, of dimension $n \geq 2$. The cost is $c(x, y) = d(x, y)^2/2$, with

$$I(x) = \{a : \exists r > 1,\ d(x, \exp_x(ra)) = r|a|_x\}, \qquad G_x = \{a : d(x, \exp_x a) = |a|_x\}.$$

The former is open; the latter is the closed minimizing domain. The source's weak MTW hypothesis is exactly

$$\mathfrak{S}_{(x,a)}(\xi, \eta) = -\frac{3}{2}\partial_s^2\partial_t^2 c(\exp_x(t\xi), \exp_x(a + s\eta))\big|_{s=t=0} \geq 0 \quad (a \in I(x),\ \langle \xi, \eta\rangle_x = 0). \tag{1.1}$$

It assumes neither nonfocality nor prior convexity. A potential has the negative-cost convention $u(x) = \max_y\{-c(x, y) - v(y)\}$, with continuous datum and, when required, its dual completion. Every minimizing log of every active target is retained. The source asserts that every ordinary semiconvex subgradient $p \in \partial u(x)$ is in G_x and satisfies

$$u(x') \geq u(x) + c(x, \exp_x p) - c(x', \exp_x p) \quad (x' \in M). \tag{1.2}$$

It also asserts that $F_t(x, p) = \exp_x(tp)$ is a homeomorphism from the whole subgradient graph to M for $0 < t < 1$, but not generally at $t = 1$. Convexity of $I(x)$ and G_x follows in its argument. These are source claims, not conclusions certified by this report's finite tests.

1.1 Complete architecture and selected interface

All five scientific modules were freshly read, including the optional degree argument. Their roles are as follows.

1. `preliminaries.tex` establishes shortening $rG_x \subset I(x)$, the smooth nonconjugate branch Hessian, the index-form minimum, strict radial comparison and transverse MTW concavity.

2. `conjugacy.tex` uses a transported zero-endpoint Jacobi field to obtain $A(a)(\xi, \xi) \leq C_0 - c_0/P(a)$. It then combines transverse concavity, the radial identity and a joint tangential–radial estimate to exclude conjugacy in a minimizing finite convex hull with nonconjugate vertices. It does not invert the exponential map at the singular point.

3. `geometry.tex` proves active-log hull representations, short-time graph coordinates and capture of every minimizing log. At a hypothetical collision it cancels linear terms at the finite stage, before dividing by the squared collision distance. This is not a finite-target assumption on the potential. Invariance of domain and a one-sheeted covering argument then globalize the continuation.

4. The same module obtains convex gap sections, intermediate $C^{1,1}$ regularization and pole control. Only support and minimizing velocities pass to time one.

5. `appendices/degree.tex` gives an optional route: local growth and openness, the sign of the graph Jacobian, degree one and the area formula. Local matrix signs alone do not supply these topological inputs.

Our interface is the divided Hessian in geometry lines139–185 and the trial-field argument in conjugacy lines48–98. The source already proves strict radial comparison and the scalar singular bound; neither is credited here as a new result. The increment is an explicit full-kernel residue, a finite-radius norm bound and a compatible-data error budget.

2 The physical endpoint problem

Fix $a \in G_x$, $a \neq 0$, and the constant-speed minimizing geodesic $\gamma(t) = \exp_x(ta)$, $0 \le t \le 1$. Use parallel orthonormal frames along this one geodesic. The curvature matrix $\mathcal{R}(t)u = R(u, \dot\gamma)\dot\gamma$ is real symmetric, and

$$Q_T(X,Y) = \int_0^T \left(\langle X', Y'\rangle - \langle \mathcal{R}X, Y\rangle\right) dt, \qquad \mathcal{L}X = -X'' - \mathcal{R}X. \tag{2.1}$$

The measure is dt for the source's time parameter, not arclength silently rescaled. Tangential fields are included. On $H^1_0(0,T;\mathbb{R}^n)$, $Q_T \ge 0$ by the second variation of minimizing energy. With smooth coefficients, the Dirichlet realization of $\mathcal{L}$ has domain $H^2 \cap H^1_0$ in $L^2(dt;\mathbb{R}^n)$ and compact resolvent.

Let C, S solve

$$C'' + \mathcal{R}C = 0, \quad C(0) = \mathrm{Id},\ C'(0) = 0; \qquad S'' + \mathcal{R}S = 0, \quad S(0) = 0,\ S'(0) = \mathrm{Id}. \tag{2.2}$$

For $0 < T < 1$, $S(T)$ is invertible by proper shortening. The field with $J(0) = \xi$, $J(T) = 0$ is

$$J(t) = C(t)\xi - S(t)S(T)^{-1}C(T)\xi.$$

Integration by parts and the index minimum give the source's exact normalization

$$H_T := \frac{1}{T}A_x(Ta) = S(T)^{-1}C(T), \qquad \langle H_T\xi, \xi\rangle = Q_T(J,J) = -\langle \xi, J'(0)\rangle. \tag{2.3}$$

Here $A_x(Ta)$ differentiates the initial point while keeping the second endpoint fixed. Thus H_T, not $A_x(Ta)$, is the duration-T index operator. The Wronskians imply

$$S^TS' = S'^TS, \quad C^TS' - C'^TS = \mathrm{Id}, \quad H_T^T = H_T, \quad H_T' = -S(T)^{-1}S(T)^{-T}. \tag{2.4}$$

For example, the last identity follows by differentiating $S^{-1}C$ and using $C' - S'S^{-1}C = -S^{-T}$. These are classical identities already used by the source, not finite-dimensional substitutes for geodesic minimality.

3 Compatibility exactly at a conjugate endpoint

Suppose $S(1)$ is singular. Let $V : \mathbb{R}^k \to \mathbb{R}^n$ have orthonormal columns spanning $\ker S(1)$, and put

$$D = S'(1)V, \qquad J_v(t) = S(t)Vv. \tag{3.1}$$

Uniqueness of the Jacobi initial-value problem makes D injective: $J_v(1) = J_v'(1) = 0$ would imply $v = 0$. The Wronskians give

$$D^TS(1) = 0, \qquad D^TC(1) = V^T. \tag{3.2}$$

The first identity and equality of nullities show $\operatorname{ran} D = \ker S(1)^T$. Notice that this is generally not $\ker S(1)$ in the parallel frame.

Theorem 3.1 (Terminal compatibility, not a singular inverse)**.** *For a prescribed initial displacement $\xi \in \mathbb{R}^n$, the endpoint problem $\mathcal{L}J = 0$, $J(0) = \xi$, $J(1) = 0$ is solvable if and only if*

$$V^T \xi = 0. \tag{3.3}$$

When solvable, its solutions differ by $S(t)Vv$, and all have the same index value. That value is the minimum of $Q_1(X, X)$ over $X \in H^1$, $X(0) = \xi$, $X(1) = 0$. If (3.3) fails, the infimum on that affine space is $-\infty$.

Proof. Every Jacobi field with the initial value has form $C(t)\xi + S(t)b$. The terminal equation is $S(1)b = -C(1)\xi$. Its exact left-null-space compatibility condition is $D^T C(1)\xi = 0$, which is (3.3). The freedom is $b \mapsto b + Vv$. If J is one solution, every admissible X is $J + Z$ with $Z \in H_0^1$. Integration by parts gives $Q_1(J, Z) = 0$, so $Q_1(X, X) = Q_1(J, J) + Q_1(Z, Z) \geq Q_1(J, J)$. A kernel field changes neither this minimum nor $-\langle \xi, J'(0)\rangle$, since $\xi \perp \operatorname{ran} V$. If compatibility fails, choose v with $\langle \xi, Vv\rangle \neq 0$ and any affine lift E of $(\xi, 0)$. Then $Q_1(J_v, J_v) = 0$ and $Q_1(E, J_v) = -\langle \xi, Vv\rangle$. The index of $E + tJ_v$ is an affine function of t with nonzero slope, hence is unbounded below. □

This does not contradict nonnegativity on endpoint-zero fields: the incompatible affine class has a moving initial endpoint. A Moore–Penrose solution of an incompatible boundary row is only a least-squares endpoint fit. It is not a Jacobi field satisfying the prescribed traces.

4 A rescaled matrix and an explicit pole remainder

Complete V to an orthogonal matrix $[V, W]$. Since the radial direction is not conjugate, $1 \leq k < n$. The ranges of D and $S(1)W$ are orthogonal by (3.2), and the latter map is injective. Define the positive constants

$$\delta = \sigma_{\min}(D), \quad \sigma = \sigma_{\min}(S(1)W), \quad g_0 = \min(\delta, \sigma). \tag{4.1}$$

Choose $0 < \varepsilon_0 \leq 1$. On $[1 - \varepsilon_0, 1]$ suppose verified bounds are available:

$$\|S'\| \leq L, \quad \|S''\| \leq 2h, \quad \|C\| \leq c, \quad \|C'\| \leq c_1, \qquad \nu = h + L. \tag{4.2}$$

All are physical Euclidean operator norms in these orthonormal frames. Set

$$B_\varepsilon = [S(1-\varepsilon)V/\varepsilon,\ S(1-\varepsilon)W], \quad B_0 = [-D,\ S(1)W], \quad T_\varepsilon = [V, \varepsilon W]. \tag{4.3}$$

B_0 is invertible, but $S(1)$ is not. These statements concern different unknown scalings.

Theorem 4.1 (Rank-resolved pole certificate)**.** *Under the preceding assumptions, for $0 < \varepsilon \leq \varepsilon_0$ with $\nu\varepsilon \leq g_0/2$, define*

$$\mathcal{P} = V(D^T D)^{-1} V^T, \qquad K_0 = \frac{2c}{g_0}\left(1 + \frac{\nu}{g_0}\right) + \frac{c_1}{g_0}. \tag{4.4}$$

Then

$$\|B_\varepsilon^{-1}\| \leq \frac{1}{g_0 - \nu\varepsilon}, \qquad \|H_{1-\varepsilon} + \varepsilon^{-1}\mathcal{P}\| \leq K_0. \tag{4.5}$$

Consequently the residue is negative semidefinite in H, has rank k, and is determined by both the initial kernel and the physical final derivatives of its Jacobi fields.

Proof. Taylor's formula on the kernel columns and the fundamental theorem of calculus on the complementary columns give

$$\|S(1-\varepsilon)V/\varepsilon + D\| \le h\varepsilon, \qquad \|(S(1-\varepsilon) - S(1))W\| \le L\varepsilon.$$

Thus $\|B_\varepsilon - B_0\| \le \nu\varepsilon$. Orthogonality of the ranges in B_0 gives $\sigma_{\min}(B_0) = g_0$. The perturbation inequality for the smallest singular value proves the first bound, and the resolvent identity gives

$$\|B_\varepsilon^{-1} - B_0^{-1}\| \le \frac{\nu\varepsilon}{g_0(g_0 - \nu\varepsilon)}.$$

From $S(1-\varepsilon)[V, W] = B_\varepsilon \operatorname{diag}(\varepsilon \mathrm{Id}_k, \mathrm{Id}_{n-k})$ we have the exact identity

$$\varepsilon H_{1-\varepsilon} = T_\varepsilon B_\varepsilon^{-1} C(1-\varepsilon). \tag{4.6}$$

The first block row of B_0^{-1} is $-(D^T D)^{-1} D^T$. Therefore

$$T_0 B_0^{-1} C(1) = -V(D^T D)^{-1} D^T C(1) = -\mathcal{P}.$$

Since $\|T_0\| = 1$, $\|T_\varepsilon - T_0\| = \varepsilon$ and $\|C(1-\varepsilon) - C(1)\| \le c_1\varepsilon$, subtracting this identity from (4.6) gives

$$\|\varepsilon H_{1-\varepsilon} + \mathcal{P}\| \le \varepsilon \left\{ \frac{c}{g_0 - \nu\varepsilon} + \frac{\nu c}{g_0(g_0 - \nu\varepsilon)} + \frac{c_1}{g_0} \right\} \le K_0\varepsilon.$$

Divide by ε. No inverse was assigned to $S(1)$. □

Remark 4.2 (Basis and normalization). For any other, possibly nonorthogonal, kernel basis VZ with invertible Z, the expression $VZ[(DZ)^T(DZ)]^{-1}(VZ)^T$ equals $\mathcal{P}$. Replacing the middle Gram matrix by the identity destroys this invariance. The bounded remainder theorem concerns $H_T = A_x(Ta)/T$; multiplying by T yields the corresponding branch Hessian and changes its bounded part. It does not define the true-cost Hessian at a cut point.

Corollary 4.3 (Kernel contamination scale)**.** *For every ξ, put $q = \|V^T\xi\|$. Under Theorem 4.1,*

$$-\frac{q^2}{\delta^2\varepsilon} - K_0\|\xi\|^2 \le \langle H_{1-\varepsilon}\xi, \xi\rangle \le -\frac{q^2}{\|D\|^2\varepsilon} + K_0\|\xi\|^2. \tag{4.7}$$

For a bounded family ξ_ε, its index values are bounded in absolute value if and only if $\|V^T\xi_\varepsilon\| = O(\sqrt{\varepsilon})$. If $\xi \perp \operatorname{ran} V$ is fixed, its index values tend to the terminal minimum in Theorem 3.1.

Proof. The eigenvalues of $(D^T D)^{-1}$ lie between $\|D\|^{-2}$ and δ^{-2}; apply (4.5). This proves the equivalence, including its necessity from the upper bound. For compatible fixed ξ, (4.5) bounds $H_{1-\varepsilon}\xi$ as a vector. Hence the Jacobi initial derivatives $-H_{1-\varepsilon}\xi$ have convergent subsequences. Continuous dependence of (2.2) shows that every limit field has terminal value zero and initial value ξ. Its index equals the unique terminal minimum, so all subsequential scalar limits coincide. □

5 Validated endpoint data and compatible residuals

The rescaling isolates a geometric singularity; it does not remove the need to validate the data. Suppose the kernel basis and its dimension are already certified. Let $\widetilde{B}, \widetilde{C}$ approximate $B_\varepsilon, C(1-\varepsilon)$ in physical operator norm with errors e_B, e_C. Write $g_\varepsilon = g_0 - \nu\varepsilon$.

Proposition 5.1 (Finite endpoint enclosure)**.** *If $e_B < g_\varepsilon$, then $\widetilde{B}$ is invertible and, for $\widetilde{H}_\varepsilon = T_\varepsilon \widetilde{B}^{-1}\widetilde{C}$,*

$$\|\varepsilon H_{1-\varepsilon} - \widetilde{H}_\varepsilon\| \le \frac{e_C}{g_\varepsilon} + \frac{e_B\|\widetilde{C}\|}{g_\varepsilon(g_\varepsilon - e_B)}. \tag{5.1}$$

The error for the unscaled Hessian is this bound divided by ε.

Proof. The singular-value perturbation bound gives $\|\widetilde{B}^{-1}\| \le (g_\varepsilon - e_B)^{-1}$. Subtract the two inverse products, use $B^{-1} - \widetilde{B}^{-1} = B^{-1}(\widetilde{B} - B)\widetilde{B}^{-1}$, and $\|T_\varepsilon\| \le 1$. □

An unstructured error e_S in $S(1-\varepsilon)$ contributes as much as e_S/ε to the kernel columns of B_ε. Small fixed absolute matrix error is therefore insufficient near conjugacy. Certified Taylor remainders, interval ODE bounds or analytic model values could supply the required data; point samples and positive finite Gram matrices cannot certify the missing continuum or the exact kernel dimension.

Proposition 5.2 (Energy enclosure on the terminal compatible space)**.** *Suppose $V^T\xi = 0$, and let J be a terminal Jacobi solution. Let $X \in H^2(0,1;\mathbb{R}^n)$ have the exact traces $X(0) = \xi$, $X(1) = 0$, and set $f = \mathcal{L}X$. If the positive spectrum of the Dirichlet operator $\mathcal{L}$ is bounded below by a verified $\lambda_+ > 0$, then*

$$Q_1(X,X) - \frac{\|f\|_{L^2}^2}{\lambda_+} \le Q_1(J,J) \le Q_1(X,X). \tag{5.2}$$

The full residual is used, not only its projection on trial fields.

Proof. Put $e = X - J \in H^2 \cap H_0^1$. Then $\mathcal{L}e = f$, $f \perp \ker\mathcal{L}$, and $Q_1(X,X) - Q_1(J,J) = Q_1(e,e)$. Expand the component of e perpendicular to the kernel in the Dirichlet eigenbasis. If f_j are the positive-mode residual coefficients, then $Q_1(e,e) = \sum_j |f_j|^2/\lambda_j \le \|f\|^2/\lambda_+$. Kernel components make no contribution. This also proves the lower bound. □

Existence of some positive complement gap follows for a fixed smooth finite-interval problem by compact resolvent. Its numerical lower bound is additional input. A finite Ritz computation alone gives no such lower bound on the full positive spectrum. Approximate traces must first be lifted, with their induced interior residual retained.

6 Worked models and negative controls

Example 6.1 (A minimizing half great circle). On the unit two-sphere, take a minimizing geodesic of length π, parametrized on $[0,1]$. In the parallel tangential–normal frame,

$$\mathcal{R} = \operatorname{diag}(0,\pi^2), \quad S(t) = \operatorname{diag}(t, \sin(\pi t)/\pi), \quad C(t) = \operatorname{diag}(1, \cos(\pi t)).$$

Thus $V = (0,1)^T$, $D = (0,-1)^T$, $W = (1,0)^T$, and $g_0 = 1$. The radial direction is compatible and has terminal minimum ξ_1^2; any nonzero fixed normal displacement is incompatible. For $T = 1-\varepsilon$,

$$H_T = \operatorname{diag}(T^{-1}, \pi\cot(\pi T)), \qquad \pi\cot(\pi(1-\varepsilon)) = -\varepsilon^{-1} + \frac{\pi^2}{3}\varepsilon + O(\varepsilon^3). \tag{6.1}$$

One may take $L = 1$, $h = \pi/2$, $c = 1$, $c_1 = \pi$ and $\varepsilon_0 = 1/8$. Using $\pi < 22/7$ gives $\nu < 18/7$, so $\nu\varepsilon_0 < 1/2$, and Theorem 4.1 permits $K_0 = 72/7$. This is deliberately a simple validated upper bound, not the optimal sphere remainder. For $\xi_\varepsilon = (1, \varepsilon^b)$ with $b > 0$, the normal contribution is

$-\varepsilon^{2b-1} + O(\varepsilon^{2b+1})$. It diverges for $b < 1/2$, has finite nonzero limit for $b = 1/2$, and vanishes for $b > 1/2$.

The terminal Dirichlet spectrum is the union of $\{m^2\pi^2 : m \geq 1\}$ in the tangential direction and $\{(m^2-1)\pi^2 : m \geq 1\}$ in the normal direction. Hence $\lambda_+ = \pi^2$. For compatible initial datum $\xi = (u, 0)$ and

$$X(t) = ((1-t)u + v\sin(\pi t), 0),$$

the exact minimum is u^2 and

$$Q_1(X, X) - u^2 = \frac{\pi^2 v^2}{2}, \qquad \|\mathcal{L}X\|^2_{L^2} = \frac{\pi^4 v^2}{2}.$$

The lower enclosure in Proposition 5.2 is attained. No global MTW assertion is needed to compute this one minimizing geodesic.

Example 6.2 (A smooth variable-curvature residue). For a parameter $d > 0$, define on $[0, 1]$

$$\begin{aligned} q_d(t) &= 1 + t + (4d-5)t^2 + (3-3d)t^3, \\ j_d(t) &= t(1-t)q_d(t), \\ R_d(t) &= \frac{36 - 24d + 60(d-1)t}{q_d(t)}. \end{aligned} \tag{6.2}$$

The cubic Bernstein coefficients of q_d are $1, 4/3, 4d/3, d$, so $q_d > 0$ on the closed interval. Direct differentiation gives $j_d'' + R_d j_d = 0$, with

$$j_d(0) = j_d(1) = 0, \quad j_d'(0) = 1, \quad j_d'(1) = -d, \quad j_d''(0) = j_d''(1) = 0.$$

Thus R_d is smooth, not a singular potential obtained by illicitly dividing at an endpoint. For compactly supported test functions z, integration by parts gives the ground-state identity

$$\int_0^1 (z'^2 - R_d z^2)\, dt = \int_0^1 j_d^2 \left((z/j_d)'\right)^2 \, dt \geq 0.$$

Density extends nonnegativity to H_0^1; the nonzero Dirichlet kernel is spanned by j_d. Appending a zero-curvature tangential coordinate produces a smooth symmetric Jacobi system with nonnegative index, $S(t) = \mathrm{diag}(t, j_d(t))$, and residue

$$\mathcal{P} = \mathrm{diag}(0, d^{-2}). \tag{6.3}$$

The other scalar solution $c_d(0) = 1, c_d'(0) = 0$ obeys the Wronskian $c_d j_d' - c_d' j_d = 1$, hence $c_d(1) = -1/d$. In particular the residue is not generally the orthogonal projection onto the kernel. At $d = 2$ its normal coefficient is $1/4$, even though the initial kernel basis has norm one. This is an exact Jacobi-operator model; we do not assert that it has been embedded in a closed manifold satisfying weak MTW.

Example 6.3 (An exact local Taylor row is not an endpoint certificate). Choose $\psi \in C_c^\infty(0, 1)$, $\psi \geq 0$, not identically zero, and consider $L_A = -d^2/dt^2 - A\psi$ with Dirichlet domain. Its first eigenvalue satisfies $\lambda_1(0) = \pi^2$, and the min–max characterization gives

$$|\lambda_1(A) - \lambda_1(B)| \leq |A - B|\|\psi\|_\infty.$$

Take a nonzero test function supported where $\psi > 0$. Its Rayleigh quotient becomes negative for sufficiently large A, so some first crossing $A_* > 0$ has $\lambda_1(A_*) = 0$ and $L_{A_*} \geq 0$. A first

eigenfunction is positive inside the interval; its initial derivative cannot vanish by ODE uniqueness. After normalizing that derivative to one it is exactly the fundamental solution S and satisfies $S(1) = 0$. By contrast, $A = 0$ gives $S(t) = t$, with $S(1) = 1$.

Both curvature coefficients vanish near the initial endpoint. Their initial jets of every order, and those of the normalized fundamental solutions, therefore agree. This is a continuum counterexample to inferring endpoint invertibility from initial jets, not a counterexample to the source's geometric theorem. Adding a zero-curvature coordinate retains a tangential direction.

Example 6.4 (A projected residual misses a Jacobi error). For the normal sphere operator $L = -d^2/dt^2 - \pi^2$, let the trial test space contain the first $m-1$ sine modes. The field $z(t) = \sin(m\pi t)$, $m \geq 2$, has zero projected residual, but

$$Lz = (m^2 - 1)\pi^2 z, \qquad Q_1(z, z) = \frac{(m^2 - 1)\pi^2}{2} > 0.$$

Projection on a physically orthonormal finite set still misses the full residual. Nonorthogonal coefficients add a metric issue; they do not remove this infinite-dimensional one.

7 Connection to the original Ram boundary proofs

Fresh original reading of TR212 [6], lines178–306, includes its actual differentiated ODE recurrence, Taylor trace rows and left-null-space compatibility proof. For the matrix Jacobi equation, the same Leibniz identity becomes

$$J^{(r+2)}(t_0) = -\sum_{j=0}^{r} \binom{r}{j} \mathcal{R}^{(j)}(t_0) J^{(r-j)}(t_0). \tag{7.1}$$

It is an initial-value recurrence, not a two-endpoint inverse. For a smooth solution its order-N Taylor trace at $t_0 + h$ carries the full remainder bounded by

$$\frac{|h|^{N+1}}{(N+1)!} \sup_{t \text{ between } t_0 \text{ and } t_0+h} \|J^{(N+1)}(t)\|.$$

Derivative traces have their corresponding lower-order remainders. Example 6.3 explains why no finite or infinite initial jet alone supplies these bounds for an arbitrary smooth coefficient.

TR212 lines430–475 prove the need for boundary lifting and present the resonant Dirichlet compatibility obstruction. Lines759–808 state the residual-to-error theorem conditional on an inverse in specified spaces, including boundary residuals. Theorem 3.1 identifies that obstruction concretely for the source's Jacobi problem. At conjugacy we use the positive complement only; before conjugacy Theorem 4.1 isolates the precise singular scaling instead of assuming a uniform inverse. This is a source-specific application of that original boundary logic, not a claim that the local recurrence proves global support.

The physical-metric chapter [7], freshly read in full, proves that a synthesis J carries the Gram matrix $G = J^*J$, and that a projected residual omits the orthogonal complement. Its exact metric and residual identities justify two concrete requirements here: the terminal derivative Gram matrix $D^T D$ in (4.4) must be retained, and the residual in (5.2) is the complete $L^2(dt)$ residual. We read the chapter's proofs and examples, not the external reports it cites or its included illustration file. Parallel orthonormal frames are an actual isometry of the geodesic bundle, not an arbitrary coefficient normalization.

8 What the local certificate does and does not change

The source's one-field trial provides a one-sided blow-up estimate $C_0 - c_0/P$ when a genuine conjugacy kernel is hypothesized. Our result resolves all kernel directions simultaneously, fixes the leading physical coefficient and bounds the remaining matrix in norm. It also quantifies how accurately a boundary direction must avoid that kernel. The finite endpoint enclosure is useful for a validated ODE calculation because it names the scaled matrix, all norm errors and their amplification. These refinements, with full local proofs and nontrivial variable-curvature examples, support the local C3 assessment. The Fredholm alternative, Wronskians, singular-value perturbation and spectral residual argument are classical; no historical priority for those tools is claimed.

There are concrete costs. The constants deteriorate with g_0, so the certificate is not uniform over arbitrary metrics, geodesics or changing kernel multiplicity. The exact kernel and the positive complement gap are separate validation tasks. Unstructured endpoint matrix errors are amplified by $1/\varepsilon$ in forming the scaled columns and again when recovering the unscaled Hessian. The theorem supplies no automatic computational speed advantage.

The result is not a proof of the source's conjugacy exclusion for a velocity segment: that proof also uses transverse MTW concavity and joint radial–tangential control, not merely the pole along one fixed ray. In particular, constants for this fixed geodesic may not be substituted uniformly along a changing family without proof. Nor does the endpoint certificate prove the arbitrary-potential local collision lemma. The source cancels its active barycenter terms exactly before division by the squared collision distance. An approximate barycenter error e leaves a term of size $O(\ell e)$; a uniform fixed $e > 0$ is not negligible relative to ℓ^2 as $\ell \downarrow 0$. One needs $e = o(\ell)$ or a declared positive distance cutoff and error budget.

Invariance of domain, the homotopy of the entire compact subgradient graph, capture of every minimizing log, and the covering or optional degree argument remain global obligations. At time one, a potential $u(x) = -c(x, y_0)$ can send many poles to y_0. No local Hessian estimate establishes injectivity or $C^{1,1}$ regularization there.

8.1 Conditional extensions and future work

The local analysis needs a smooth symmetric Jacobi system, no earlier conjugate time, and the relevant index nonnegativity; weak MTW is not used in its algebra. It therefore applies to any independently verified system with these properties, but says nothing by itself about global cost support for that system. A useful next step is a validated ODE scheme that bounds S, S', S'', C, C' and the kernel/complement data in the physical metric. A second is a uniform family version with controlled kernel rank, lower g_0 and a parameter-dependent remainder. Kernel-rank changes require a different argument. A third is a joint tangential and radial residue theorem compatible with the source's minimizing convex hulls; a collection of separate fixed-ray limits is not enough.

9 Public status and actual verification scope

The October9,2026 public checks distinguished source declarations from external mathematical evidence. Figalli–Gallouet–Rifford's published Theorem1.7 explicitly retains the nonfocal hypothesis [2]; its abstract, introduction and theorem statement were inspected, not its complete proof. Cordero-Erausquin–McCann–Schmuckenschlager's Claim2.4 and Proposition2.5, including the endpoint-moving index-form proof, were inspected in the author-hosted published article [3]. That classical predecessor is also explicitly credited by the assigned source. The source's all-dimensional weak-MTW global theorem remains attributed to its preprint [1].

A September2026 paper on Hamilton–Jacobi cut loci [4] was checked at abstract and version-record level only. Its touching and propagation results are contextual, not a replacement proof of the assigned theorem. The repository's family360 scope note [5] was read as a scope declaration. It explicitly distinguishes its finite-target scale/local injectivity result from the manuscript's arbitrary-potential local injectivity assertion. No Lean proof, axiom audit or build was performed. A fresh path-specific GitHub history query returned the initial commit `adc7f124` dated October6,2026; this is metadata, not proof validation or a substitute for hashing the cached input files.

All eight assigned public text files were freshly displayed: six TeX files, the bibliography and README, totaling 1,999 lines, with no unread text complement. Original private reads were TR212 lines178–306, 430–475 and759–808 (225 lines) and the physical-metric chapter lines1–210. The combined private total is435 lines; the remaining828 lines of TR212 are explicitly unread. The external files cited or included by those originals are not silently included in this scope. PRE hashes were persisted before the proof displays; POST hashes and exact interval complements are retained in the owned ledger.

The owned `M690-checks.py` passed 207 exact cases and 489 counted assertions, and separately eight float64 cases with eight assertions. The script evaluates actual Jacobi model identities, physical residues, endpoint compatibility, rescaled inverse perturbations and the manufactured negative controls. Exact symbolic checks and numerical diagnostics are reported separately in the final receipt, with the executed script hash and runtime. The matrix-product perturbation fixture uses an explicitly manufactured C matrix and does not claim it is the fundamental Jacobi $C(t)$ of that model. No parent cases or foreign checker results are counted. Finite checks do not verify a global manifold construction, all weak-MTW inequalities, the continuum derivative bounds, or the source's covering argument.

10 Conclusion

A singular endpoint requires compatibility, not an invented inverse. The source's proper radial contractions admit an explicit, rank-resolved pole certificate whose coefficient is the physical Gram inverse of the terminal Jacobi derivatives. This gives a useful local error and boundary contamination account while leaving global support and convexity tied to their genuine MTW and topological arguments.

References

[1] OpenAI, *Global Support and Convex Injectivity Domains under Weak MTW*, September25,2026. Pinned public manuscript. All assigned scientific text read; external and global proof inputs remain as identified in the report.

[2] A. Figalli, T. O. Gallouet and L. Rifford, *On the convexity of injectivity domains on nonfocal manifolds*, SIAM J. Math. Anal.47(2015),969–1000, author-hosted published text, Theorem1.7; selected statement and context inspected.

[3] D. Cordero-Erausquin, R. J. McCann and M. Schmuckenschlager, *A Riemannian interpolation inequality a la Borell, Brascamp and Lieb*, Invent. Math.146(2001),219–257, author-hosted published text, Claim2.4 and Proposition2.5, selected proofs inspected.

[4] P. Cannarsa, W. Cheng, J. Hong and W. Wei, *On the cut locus of Hamilton–Jacobi equations I: structure and propagation via the touching approach*, arXiv:2609.02879v1, September2,2026. Abstract and version metadata only; full proof not audited.

[5] OpenAI Math repository, family360 formalization scope declaration, accessed October9,2026. Declaration only; no Lean proof verification here.

[6] M. Subbarao, confidential original TR212, local directory `res-rt/TechReps/FinalTechReps/TRs`; subdirectory `TR212-boundary-value-ram-operators`, same-stem TeX. Fresh proof reads178–306,430–475,759–808.

[7] M. Subbarao, confidential original physical-metric chapter, `RamCalBooks/volumes/volume21/chapters`, file `ch01d-physical-metrics.tex`, lines1–210 freshly read. Included figure and external references not independently inspected.